# Clubdash Master System Architecture

Facility Web OS & Business Intelligence Matrix • Version 1.2 • Ecosystem Topology

## 1. Ecosystem Topology & Hardware Matrix

The Clubdash platform fundamentally decouples physical real estate ownership from community programming. By providing dedicated software channels for every actor in the recreation supply chain, the platform converts static venue space into a highly monetizable, fluid asset. The system serves three distinct actors operating on specific hardware environments to eliminate operational friction and ensure absolute data synchronicity.

| Actor Archetype | Role & Objective | Software Interface | Access Status |
|---|---|---|---|
| **Facility Owner** | Real Estate/Landlord. Maximizes inventory utilization and B2B rentals. | Facility Web OS & Quickdash Tablet | `[LOCKED]` (Verified Owners) |
| **Organizer / Coach** | Promoter/Talent. Manages rosters, builds funnels, sources venues. | Clubdash Pro App | `[OPEN]` (Self-Serve) |
| **Consumer / Player** | End-user. Books courts, buys event tickets, consumes media. | Sodia App | `[OPEN]` (Public) |

## 2. Engineering Infrastructure: The Agnostic Core

At the center of the Facility Web OS is the Spatial Sub-Division Engine. Unlike legacy scheduling software that treats a "Gymnasium" as a singular, indivisible asset, Clubdash understands nested spatial relationships. A primary basketball court can be programmatically divided into three sub-pickleball courts. If an Organizer books "Pickleball Court A," the system automatically locks the parent "Basketball Court" from concurrent bookings while leaving "Pickleball Courts B & C" available in the Sodia consumer feed. This agnostic core maximizes yield per square foot.

## 3. Inventory Rules & The B2B Marketplace

To support high-velocity operations, the platform utilizes dynamic time-blocks and automated cross-sport changeover buffers. Facility Owners configure default operational logic once. For example, transitioning a turf field from soccer to lacrosse automatically inserts a 15-minute buffer into the master calendar, preventing overlapping access. This logic integrates seamlessly with the B2B Marketplace. Organizers query the facility's available space, negotiate terms via digital affiliation handshakes, and execute binding rental agreements directly through the system without manual intervention.

| 98.4% | 15min | Zero |
| --- | --- | --- |
| UPTIME SYNC ACCURACY | AVG. AUTO-BUFFER | DOUBLE BOOKINGS |

## 4. Financial Routing & The Matrix

The core of the Facility OS value proposition is absolute financial transparency and automated reconciliation. Through Stripe Connect Split Ledgers, every transaction is instantly parsed at checkout. When a consumer buys a $100 tournament ticket through an Organizer's funnel on the Sodia App, the routing engine deducts the platform fee, routes the venue rental percentage directly to the Facility Owner's merchant account, and deposits the remainder to the Organizer. This eliminates end-of-month invoicing, bad debt, and manual collections.

### Dual-Membership Identity Matrix

Facility Pro Tiers allow owners to sell base facility access (Tier 0 Base Waivers) to consumers. If a consumer enters a facility for an Organizer's event but lacks the venue's base liability waiver, the system forces a digital signature and optional micro-transaction before generating the entry QR code.

## 5. Quickdash: Staff Operations & DTR

The Facility "Quickdash" interface is a bifurcated mobile and tablet application designed strictly for front-desk staff and floor managers. It replaces disparate point-of-sale systems, manual clipboards, and disparate entry gates. Crucially, the app is offline resilient. It caches active ticket lists locally; if the facility's internet drops, scanning operations continue uninterrupted, syncing logs via payload immediately upon network reconnection.

### THE FLOAT GATE & ACCOUNTABILITY ENGINE

Loss prevention is managed via the Float Gate protocol. Before a shift begins, staff must perform a Mobile Selfie Clock-In via their Sodia app, which is geofenced and timestamped to the facility. Next, they must log into the Quickdash tablet using a secure PIN and verify the Opening Float (starting cash). At shift end, a blind cash count is strictly enforced to secure the ledger, locking the drawer until the system matches the physical input against recorded sales.

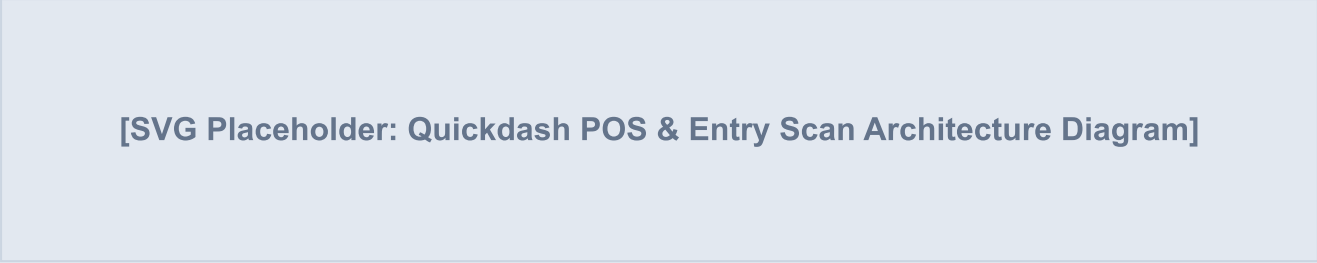



Figure 1: Quickdash offline-resilient payload synchronization cycle.

## 6. The Master Grid & Desktop Web OS

The Desktop Web OS provides the Facility Admin with complex macro-management capabilities. The centerpiece is the Master Grid—a robust X/Y calendar allowing intuitive drag-and-drop mechanics for parent/child asset bookings. It serves as the ultimate command center, tracking walk-ins, B2B wholesale agreements, and real-time CRM updates.

## 7. Business Intelligence & Revenue Protection

To ensure Facility Owners rely on Clubdash as their primary operating system, advanced intelligence modules are hardcoded into the Web OS. The Global Time-Frame Metrics Dash recalculates all command center visuals on the fly based on daily, weekly, or quarterly toggles. Central to this is the Platform ROI Card, which explicitly isolates B2C app-sourced bookings and B2B Organizer payouts to mathematically prove the platform's monetary value to the landlord on an ongoing basis.

### Squatter Detection (Gate-vs-Booking Gap)

The system actively cross-references Quickdash entry scans against active court capacity limits. If Court 1 is booked for 10 individuals, but the gate records 14 valid Tier 0 scans associated with that specific digital pass, the system flashes a "Capacity Breach" alert to floor staff, prompting immediate upsell enforcement.

### YIELD MANAGEMENT & THE HOURLY HEATMAP

The dead-hour detection engine tracks historical utilization by hour and weekday. When structural vacancies are identified, the Web OS prompts the owner to draft an "Off-Peak SKU." With one click, discounted rates are pushed instantly to the Sodia consumer feed, converting stagnant real estate into immediate cash flow.

### THE "NEEDS ATTENTION" TRIAGE

Operational fires are minimized through a ranked triage queue on the dashboard home screen. Blind-count mismatches, pending wholesale B2B affiliation requests, and at-risk dormant Organizers (who haven't booked in 30 days) float to the top. This allows owners to act directly with "Approve," "Recount," or "Win-Back Promo" actions, streamlining facility governance.